

KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

Mathematics
Year 8

WORKBOOK TEACHER COPY

Probability

Year 8 Mathematics • Workbook

Name _____ Class _____ Teacher _____

How this workbook works

Each lesson is **three pages**. Copy your teacher's **notes**, then work through **We do** (dark boxes: we solve these together; copy every step) and **You do** (your turn, one round at a time). Rate each success criterion at the end.

Page 1 of each lesson is **Easy** ●, page 2 is **Medium** ●● and page 3 is **Hard** ●●●: a harder example, mixed questions (a tag shows an earlier lesson or year) and worded problems. A question tagged **Extension** uses Stage 5 wording.

NSW Mathematics K–10 Syllabus (2022), Stage 4: MA4-PRO-C-01. We acknowledge the Bundjalung people, the Traditional Custodians of the land on which we learn.

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Cover and banner artwork: *Kingscliff High School Vision Statement* by local Indigenous artist Marni Tuala.

1 Chance and the sample space

Probability • MA4-PRO-C-01 **EASY •**

LEARNING INTENTION

We are learning to describe chance with words and with numbers from 0 to 1, list a sample space, and decide whether outcomes are equally likely.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* place an event on the probability scale from 0 to 1. ○○○
- ▸ *I can* list the sample space of a chance experiment. ○○○
- ▸ *I can* decide, with a reason, whether outcomes are equally likely. ○○○

NOTES key ideas: students copy these into their booklet

TEACHER COPY

KEY TERMS

chance experiment: an activity with an uncertain result, e.g. a spin

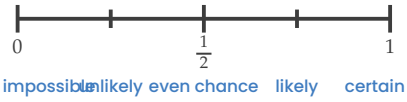
trial: one go · **outcome:** one possible result

sample space: all possible outcomes, each written once, in braces { }

equally likely: every outcome has the same chance

at random: no outcome is favoured

THE PROBABILITY SCALE



A probability is a number from 0 to 1. It can be a fraction, a decimal or a percentage: $\frac{1}{2} = 0.5 = 50\%$.

STEPS: THE SAMPLE SPACE

1. write each outcome once
2. put them in braces { }
3. count the outcomes
4. equally likely? check sizes, fairness



{red, blue}: 2 outcomes, not equally likely (red has 2 sectors)

WE DO with your teacher: copy every step into the space

Example 1 Answer each part. Where it says explain, give a reason.

a The scale

Write each letter where its event belongs on the scale. A: the sun rises tomorrow. B: rolling a 7 on a die. C: a coin lands heads.



b Sample space

The spinner is spun once. List the sample space, then count the outcomes.



c Equally likely?

The spinner is spun once. Is each colour equally likely? Explain.



d Sample space

List the sample space for the day of the week a person was born on. How many outcomes are there?

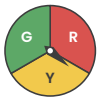
YOU DO on your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Chance words: impossible (0) · unlikely (below $\frac{1}{2}$) · even chance ($\frac{1}{2}$) · likely (above $\frac{1}{2}$) · certain (1)

ROUND 1 Questions 1–4: each spinner is spun once. Count the blue sectors and all the sectors. Then write the chance word for landing on blue.

1 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

2 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

3 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

4 What is the chance of landing on blue?



Blue sectors ___ out of ___

Chance word _____

ROUND 2 Questions 5–8: write every possible outcome once, inside braces { }. Then count how many outcomes there are.

5 A coin is tossed. List the sample space.



Sample space { _____ }

Number of outcomes _____

6 A die is rolled. List the sample space.



Sample space { _____ }

Number of outcomes _____

7 This spinner is spun. List the sample space.



Sample space { _____ }

Number of outcomes _____

8 One tile is picked. List the sample space.



Sample space { _____ }

Number of outcomes _____

Chance and the sample space

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MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER The sample space lists every possible outcome once, inside braces { }. An event is the outcomes you want. Equally likely means every outcome has the same chance.

ROUND 3 — Questions 9–11: list the sample space. If an outcome appears more than once, write it only once.

9 This spinner is spun once. List the sample space and count the outcomes.



Sample space { }

Number of outcomes

10 A person is asked the month they were born in. List the possible first letters of that month.

Sample space { }

Number of outcomes

11 Kingscliff plays a netball game. List the possible results for Kingscliff.

Sample space { }

Number of outcomes

ROUND 4 — Questions 12–14: an event is the outcomes you want. List the outcomes in each event, inside braces { }.

12 A die is rolled. List the outcomes in the event 'an odd number'.



Outcomes { }

13 One tile is picked. List the outcomes in the event 'a vowel'.



Outcomes { }

14 List the house numbers that are greater than 4 but less than 10.

Outcomes { }

ROUND 5 — Questions 15–17: decide if the outcomes are equally likely. Write *yes* or *no*, then give a reason.

15 This spinner is spun. Are red, blue and green equally likely?



Yes or no

Reason

16 A fair die is rolled. Are 1, 2, 3, 4, 5 and 6 equally likely?



Yes or no

Reason

17 Someone sits a driving test. Are 'pass' and 'fail' equally likely?

Yes or no

Reason

ROUND 6 — Questions 18–20: write the chance word that matches each probability: *impossible, unlikely, even chance, likely* or *certain*.

18 An event has a probability of **0.9**. Which chance word fits?

Chance word

19 An event has a probability of $\frac{1}{2}$. Which chance word fits?

Chance word

20 An event has a **2%** chance. Which chance word fits?

Chance word

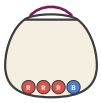
1 Chance and the sample space

Probability • MA4-PRO-C-01 HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Answer each part in a full sentence, and give the reason.

a Equally likely?
One counter is taken from the bag without looking. List the sample space. Are the outcomes equally likely?



b A match
Kingscliff plays Cabarita in a final. List the sample space of results. Are the outcomes equally likely?

c Order
A die is rolled. Put these in order from least likely to most likely: A an even number, B a 6, C a number less than 6, D a 9.



d Forecast
The forecast is a 30% chance of rain. Is rain likely or unlikely? Could it still rain?



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER 0 means impossible, $\frac{1}{2}$ means even chance, 1 means certain. List each outcome once. Equally likely means the same chance.

1 A die is rolled and a coin is tossed. Write each letter on the scale where its event belongs. A: rolling a 6. B: rolling a number less than 7. C: the coin lands heads.



2 Write $\frac{3}{4}$ as a decimal and as a percentage. Year 7

Answer

3 One tile is picked without looking. List the sample space, writing each letter once.



Answer

4 This spinner is spun once. Which colour is more likely, red or blue? Explain how you know.



Answer

5 Sam says, 'On a die, a 6 is less likely than a 1 because 6 is a bigger number.' Is Sam right? Explain.

Answer

6 A coin lands tails 7 times in a row. Is heads more likely on the next toss? Explain your answer.



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1 The forecast for the Tweed says 'a 70% chance of rain'. Write the chance word for rain. Is rain certain? Explain.

Answer

2 Colour the 4 equal sectors of this spinner so that landing on blue is *likely* but not *certain*. Then explain why your spinner works.



Answer

3 Amy and Cara are in 8K; Ben and Dev are in 8R. One of the four is picked at random to be captain. List the sample space. Is the captain more likely to come from 8K? Explain.

Answer

2 Probability of an event

Probability • MA4-PRO-C-01 **EASY**

LEARNING INTENTION

We are learning to find the probability of an event when the outcomes are equally likely, and write it as a fraction, decimal and percentage.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* count the favourable outcomes and the total outcomes. ○○○
- *I can* use P(E) = favourable ÷ total, and simplify. ○○○
- *I can* write a probability as a fraction, a decimal and a percentage. ○○○

NOTES key ideas: students copy these into their booklet — TEACHER COPY

KEY TERMS

event (E): the outcome or outcomes we want
favourable outcome: an outcome that is in the event
P(E): 'the probability of E'
simplest form: divide top and bottom by their highest common factor
a standard deck: 52 cards, 4 suits of 13 (♥ ♦ red, ♣ ♠ black); picture cards J, Q, K

FORMULA

$$P(E) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$$

fraction → decimal: top ÷ bottom
decimal → percentage: × 100%
'or' (different groups): add the counts, e.g. P(red or blue)

STEPS

1 list the outcomes 2 circle the favourable ones
3 fraction, simplify 4 convert if asked



at least 4 = 4 or more; **at most 4** = 4 or less (ext.)

$P(\text{at least 4}) = \frac{3}{6} = \frac{1}{2} = 0.5 = 50\%$

WE DO with your teacher: copy every step into the space

Example 1 Find each probability. Write it as a fraction in simplest form.

a Spinner

The spinner is spun once. Find the probability it lands on red, P(red).



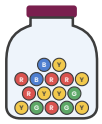
b Die

A die is rolled once. Find the probability of an odd number, P(odd).



c Lollies

One lolly is taken without looking. Find P(yellow or blue).



d Tiles

One tile is picked without looking. Find the probability of a vowel, P(vowel).

K	I	N	G	S	C	L	I	F	F

YOU DO on your own: work down the box, one round at a time. Show your working. — 8 QUESTIONS

REMEMBER Probability = $\frac{\text{favourable outcomes}}{\text{total outcomes}}$. Simplify: divide the top and bottom by the same number.

ROUND 1 Questions 1–4: each spinner is spun once. Count the blue sectors and all the sectors. Then write P(blue) as a fraction in simplest form.

1 Find the probability of landing on blue. Simplify.



Blue sectors ___ out of ___
P(blue) = _____

2 Find the probability of landing on blue. Simplify.



Blue sectors ___ out of ___
P(blue) = _____

3 Find the probability of landing on blue. Simplify.



Blue sectors ___ out of ___
P(blue) = _____

4 Find the probability of landing on blue. Simplify.



Blue sectors ___ out of ___
P(blue) = _____

ROUND 2 Questions 5–8: a fair die is rolled once. List the outcomes that count (favourable outcomes), then find the probability.

5 Find the probability of rolling a 5.



Favourable outcomes { _____
Probability _____

6 Find the probability of rolling an even number.



Favourable outcomes { _____
Probability _____

7 Find the probability of rolling a number less than 3.



Favourable outcomes { _____
Probability _____

8 Find the probability of rolling a factor of 6.



Favourable outcomes { _____
Probability _____

2 Probability of an event

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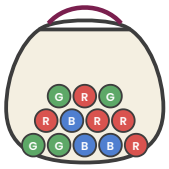
MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER Count every counter, lolly or card. 'Or' means add the counts. Decimal: top ÷ bottom. Percentage: × 100%.

ROUND 3

Questions 9–11: one counter is taken from this bag without looking. Find each probability as a fraction in simplest form.



5 red, 3 blue and 4 green counters

9 Find the probability the counter is red, P(red).

Favourable outcomes _____

Total outcomes _____

Probability _____

10 Find the probability the counter is blue, P(blue).

Favourable outcomes _____

Total outcomes _____

Probability _____

11 Find the probability the counter is red or green.

Favourable outcomes _____

Total outcomes _____

Probability _____

ROUND 4

Questions 12–14: find each probability. Write it as a fraction, then as a decimal, then as a percentage.

12 A coin is tossed. Find the probability of tails.



Fraction _____

Decimal _____

Percentage _____

13 This spinner is spun once. Find the probability of red.



Fraction _____

Decimal _____

Percentage _____

14 One tile is picked without looking. Find the probability of an F.



Fraction _____

Decimal _____

Percentage _____

ROUND 5

Questions 15–17: one card is taken from a standard deck without looking. Find each probability as a fraction in simplest form.



52 cards: 4 suits of 13. Hearts and diamonds are red. J, Q, K are picture cards (shaded).

15 Find the probability of an ace.

Favourable outcomes _____

Probability _____

16 Find the probability of a heart.

Favourable outcomes _____

Probability _____

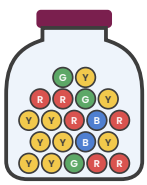
17 Find the probability of a red card.

Favourable outcomes _____

Probability _____

ROUND 6

Questions 18–20: one lolly is taken from this jar without looking. Write each answer in the form asked.



6 red, 3 green, 9 yellow, 2 blue: 20 lollies

18 Find the probability of a green lolly. Write it as a decimal.

Working _____

Answer _____

19 Find the probability of a yellow lolly. Write it as a percentage.

Working _____

Answer _____

20 Find the probability of a red or blue lolly. Write it as a fraction in simplest form.

Working _____

Answer _____

2 Probability of an event

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HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Give each answer in the form the question asks for: fraction, decimal or percentage.

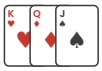
a As a decimal

One coin falls out of the money box. Find the probability it is a gold coin ($1 or $2). Write it as a decimal.



b Cards

One card is taken from a 52-card deck. Find the probability it is a red picture card (J, Q or K).



c As a percentage

A die is rolled. Find the probability of a number greater than 4. Write it as a percentage.



d Notes

Elena has two $5, four $10 and three $20 notes. She takes one without looking. Find the probability it is a $20 note.

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER Probability = favourable ÷ total. Picture cards are J, Q and K. Give the answer in the form the question asks for.

1 One jelly bean is taken from the bag without looking. Find the probability it is green or black.



Answer

2 Write $\frac{3}{8}$ as a decimal and as a percentage. Year 7

Answer

3 A whole number from 1 to 15 is chosen at random. List the prime numbers, then find the probability of a prime number.



Answer

4 One tile is picked without looking. Find the probability of an S.



Answer

5 A die is rolled. Find the probability of rolling at least 3 (that is, 3 or more). Extension



Answer

6 500 raffle tickets are sold at the school fete. Mia buys 8. Find the probability that Mia wins first prize. Write it as a percentage.



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1 An Esky has 8 lemonade, 5 orange and 2 lime cans. (a) Find the probability of taking a lemonade. (b) How many cola cans must be added so that P(lemonade) = $\frac{1}{2}$?

Answer

2 Malik has 20 cards: five 5s, four 6s, eight 7s and three 8s. Lara takes one without looking. (a) Write P(a 7) as a fraction, a decimal and a percentage. (b) Find P(an even number).

Answer

3 The probability that a day in April is rainy is 30%. April has 30 days. About how many rainy days should we expect in April?



Answer

3 Complementary events

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EASY •

LEARNING INTENTION

We are learning to identify the complement of an event, and use $P(\text{not E}) = 1 - P(E)$ to solve problems.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* write the complement of an event. ○○○
- ▸ *I can* show that $P(E) + P(\text{not E}) = 1$. ○○○
- ▸ *I can* use $P(\text{not E}) = 1 - P(E)$ with fractions, decimals and percentages. ○○○

NOTES key ideas: students copy these into their booklet

TEACHER COPY

KEY TERMS

complement of E ('not E'): every outcome in the sample space that is *not* in E

complementary events: E and not E. Together they make the whole sample space, with no overlap.

Careful: the complement of 'winning' is 'not winning' (losing *or* drawing), not just 'losing'.

FORMULA

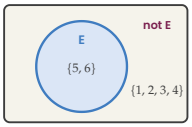
$$P(E) + P(\text{not E}) = 1$$

$$P(\text{not E}) = 1 - P(E)$$

with percentages: $P(\text{not E}) = 100\% - P(E)$

The probabilities of **all** the outcomes add to 1.

DIAGRAM



die: $E = \{5, 6\}$

$P(E) = \frac{2}{6} = \frac{1}{3}$

$P(\text{not E}) = 1 - \frac{1}{3} = \frac{2}{3}$

WE DO with your teacher: copy every step into the space

Example 1 Write the complement (the 'not' event), then find the probabilities.

a Complement

A die is rolled. Write the complement of 'rolling a 6' and list its outcomes.



b Cards

A pack has 10 red, 6 yellow and 4 green cards. One card is taken without looking. Find P(yellow) and P(not yellow).

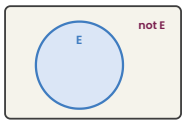
c Decimal

The probability of rain is 0.15. Find the probability of no rain.



d Diagram

A die is rolled. E = 'an even number'. Write the outcomes in the diagram, then find P(not E).



YOU DO on your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER The complement of E is 'not E': every other outcome. $P(E) + P(\text{not E}) = 1$, so $P(\text{not E}) = 1 - P(E)$.

ROUND 1 Questions 1–4: the complement of an event is everything else that could happen. Write the complement of each event.

1. A coin is tossed. Write the complement of 'tossing tails'.

   Complement _____

2. A die is rolled. Write the complement of 'rolling a 6'.

   Complement _____

3. Write the complement of 'tomorrow is a cloudy day'.

   Complement _____

4. A box has white and dark chocolates. Write the complement of 'a white chocolate'.

   Complement _____

ROUND 2 Questions 5–8: each spinner is spun once. Find P(blue) and P(not blue). Then check that the two add to 1.

5. Find P(blue) and P(not blue).



P(blue) = _____

P(not blue) = _____

Check ___ + ___ = 1

6. Find P(blue) and P(not blue).



P(blue) = _____

P(not blue) = _____

Check ___ + ___ = 1

7. Find P(blue) and P(not blue).



P(blue) = _____

P(not blue) = _____

Check ___ + ___ = 1

8. Find P(blue) and P(not blue).



P(blue) = _____

P(not blue) = _____

Check ___ + ___ = 1

3 Complementary events

Probability • MA4-PRO-C-01

MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER $P(\text{not E}) = 1 - P(E)$. With percentages: $100\% - P(E)$. Decimals: $1 - 0.15 = 0.85$. Fractions: $1 - \frac{1}{4} = \frac{3}{4}$.

ROUND 3

Questions 9–11: a fair die is rolled once. List the outcomes that are NOT the given number(s), then find the probability.

9 Find the probability of not rolling a 3.



Outcomes { }

Probability

10 Find the probability of not rolling an even number.



Outcomes { }

Probability

11 Find the probability of not rolling a prime number.



Outcomes { }

Probability

ROUND 4

Questions 12–14: use P(not E) = 1 - P(E). Give your answer in the same form as the question (percentage, decimal or fraction).

12 The probability of rain this weekend is 85%. Find the probability of no rain.

Working

Answer

13 The probability that a carton has a broken egg is 0.1. Find the probability that it has no broken eggs.

Working

Answer

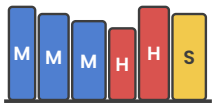
14 The probability of choosing a jack from a deck is $\frac{1}{13}$. Find the probability of not choosing a jack.

Working

Answer

ROUND 5

Questions 15–17: one book is taken from this shelf without looking. Find each probability as a fraction.



3 maths (M), 2 history (H) and 1 sport (S) book.

15 Find the probability that it is a history book.

Working

Answer

16 Find the probability that it is not a history book.

Working

Answer

17 Find the probability that it is not a science book.

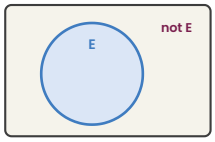
Working

Answer

ROUND 6

Questions 18–20: show an event and its complement in a diagram, and use the fact that all the probabilities add to 1.

18 A number from 1 to 10 is chosen. E = 'a multiple of 3'. Write the outcomes of E inside the circle and the outcomes of not E outside it.



19 Use your diagram from Question 18. Find P(E) and P(not E).

Working

Answer

20 Kingscliff's netball team can win, draw or lose. P(win) = 0.6 and P(draw) = 0.1. Find P(lose).

Working

Answer

3 Complementary events

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Use P(not E) = 1 - P(E). Give each answer in the same form as the question.

a Percentage

A golfer sinks 74% of putts. Find the probability that the golfer misses a putt.

b Missing value

A bag holds only red, blue and white toy cars. P(red) = 0.4 and P(white) = 0.25. Find P(blue).

c Choose

Which is the complement of 'winning a race'? A coming last B coming second C not winning D coming 2nd or 3rd

d Raffle

1000 raffle tickets are sold and Tahnee buys 5. Find the probability that Tahnee does not win.



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 6 QUESTIONS

REMEMBER P(not E) = 1 − P(E). 'Not blue and not black' means any colour except blue or black. All the probabilities add to 1.

1. A car park has 450 cars, 100 motorbikes and 10 buses. One vehicle is picked at random. Find the probability that it is not a car.

Answer

2. Work out: $1 - \frac{3}{8}$, then 1 − 0.37, then 100% − 64%. Year 7

Answer

3. A jar has 40 red, 25 blue, 50 black and 15 white jelly beans. One is taken without looking. Find the probability that it is not white.

Answer

4. Use the jar in Question 3. Find the probability that the jelly bean is not blue and not black.

Answer

5. A mobile phone number is picked at random. Find the probability that its last digit is not 0 or 1. Write it as a decimal.

Answer

6. One tile is picked without looking. Find the probability that it is not a vowel.

P R O B A B I L I T Y

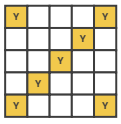
Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1. A bag holds only red, blue and white toy cars. P(red) = 0.4 and P(white) = 0.25. (a) Find P(blue). (b) The bag holds 40 cars. How many of each colour are there?

Answer

2. In this game, you win if your coin lands in a yellow square of the 5 × 5 grid. Throws that miss the grid are taken again. Find P(winning) and P(not winning).



Answer

3. Sue and Liam are in Year 8; Emily and Matt are in Year 9. One of the four is picked at random to be captain. Find P(not Emily), P(the captain is in Year 8) and P(the captain is the teacher).

Answer

4 Probability problems

Probability • MA4-PRO-C-01

EASY •

LEARNING INTENTION

We are learning to choose and use the right probability skill to solve problems from lists, tables and words.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* decide which skill a problem needs. ○○○
- *I can* set out working with P(E) = favourable ÷ total. ○○○
- *I can* explain my answer in a sentence. ○○○

NOTES key ideas: students copy these into their booklet

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PROBLEM-SOLVING CHECKLIST

1. What is the experiment? Count the **total**.
2. Which outcomes are **favourable**?
3. Is it 'not'? Use 1 − P(E).
4. Answer in the **form asked**: fraction, decimal or %.
5. **Check**: between 0 and 1? Sensible?

KEY IDEAS

'or' (different groups): add the counts

equally likely? Count each object, not each name: SUCCESS has 3 S tiles.

no memory: a coin or die does not remember earlier results

how many? number ≈ P × total, e.g. P(red) = 0.5 in a bag of 10: about 5 red

EXAMPLE



Balls numbered 1 to 20. P(a multiple of 5):

{5, 10, 15, 20}: $\frac{4}{20} = \frac{1}{5} = 0.2 = 20\%$

WE DO with your teacher: copy every step into the space

Example 1 Use the checklist in your notes. Write each probability as a fraction in simplest form, unless the question asks for a percentage.

a A table

28 students on a trip were born in these countries. One is picked at random. Find P(born in Asia).

Australia	17
Italy	3
Vietnam	4
Japan	1
China	3

b Letters

One tile is picked without looking. List the sample space. Is each letter equally likely? Explain.

S U C C E S S

c Percentage

The spinner is spun once. Find the probability of an even number. Write it as a percentage.



d Not

Alex takes one sock without looking from 2 black, 2 blue and 2 red socks. Find the probability that it is not red.



YOU DO on your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Probability = favourable outcomes ÷ total outcomes. To change a fraction to a percentage, multiply by 100%. Simplify fractions.

ROUND 1 Questions 1–4: one ball is picked at random from a barrel of 100 balls numbered 1 to 100. Find each probability as a fraction in simplest form.

1 Find the probability of the ball numbered 12.

Favourable outcomes _____

Probability _____

2 Find the probability of a number greater than 40.

Favourable outcomes _____

Probability _____

3 Find the probability of an even number.

Favourable outcomes _____

Probability _____

4 Find the probability of a number from 1 to 25.

Favourable outcomes _____

Probability _____

ROUND 2 Questions 5–8: the spinner has 5 equal sectors numbered 1 to 5 and is spun once. Write each probability as a fraction, then as a percentage.

5 Find the probability of a 2.



Fraction _____

Percentage _____

6 Find the probability of an even number.



Fraction _____

Percentage _____

7 Find the probability of a number less than 5.



Fraction _____

Percentage _____

8 Find the probability of a number from 1 to 5.



Fraction _____

Percentage _____

4 Probability problems

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MEDIUM ••

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 12 QUESTIONS

REMEMBER Checklist: count the total, count the favourable outcomes, use $1 - P(E)$ for 'not', answer in the form asked, then check. 'Or' means add the counts.

ROUND 3 Questions 9–11: one student from the trip is picked at random. Use the table. Find each probability as a fraction in simplest form.

Born in	Australia	Italy	Vietnam	Japan	China
Students	17	3	4	1	3

28 students went on the school trip to Japan.

9 Find the probability that the student was born in Australia.

Working _____

Answer _____

10 Find the probability that the student was not born in Vietnam.

Working _____

Answer _____

11 Find the probability that the student was born in Italy or China.

Working _____

Answer _____

ROUND 4 Questions 12–14: Alex takes one sock without looking from a drawer with 2 black, 2 blue and 2 red socks.

12 List the sample space (the possible colours).

Answer _____

13 Find the probability of a blue sock.

Answer _____

14 Find the probability of a pink sock.

Answer _____

ROUND 5 Questions 15–17: a whole number from 1 to 15 is chosen at random. List the favourable outcomes, then find the probability.

15 Find the probability of an odd number.

Favourable outcomes { } _____

Probability _____

16 Find the probability of a number that is not prime. (1 is not prime.)

Favourable outcomes { } _____

Probability _____

17 Find the probability of a factor of 12.

Favourable outcomes { } _____

Probability _____

ROUND 6 Questions 18–20: the 7 letters of SUCCESS are written on 7 cards. One card is picked without looking. Find each probability.

S	U	C	C	E	S	S

18 Find the probability of a vowel.

Working _____

Answer _____

19 Find the probability that the letter is not a C.

Working _____

Answer _____

20 Find the probability that the letter is also in the word FAIL.

Working _____

Answer _____

4 Probability problems

Probability • MA4-PRO-C-01 HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Harder problems: read each one twice, decide what you need to find, then solve.

a Small chances

A charity raffle sells 100 000 tickets and Sophia buys 4. Find the probability that Sophia wins. Write it as a decimal.



b Or

A whole number from 1 to 15 is chosen at random. Find the probability of a multiple of 3 or a multiple of 5.



c How many?

A bag of lollies has P(red) = 0.5 and P(green) = 0.2. The rest are purple. Find P(purple). Then find how many of each colour are in a bag of 10 lollies.

d No memory

A coin lands tails 7 times in a row. Find the probability of tails on the next toss.



YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. 6 QUESTIONS

REMEMBER $P(\text{not E}) = 1 - P(\text{E})$. Each toss or roll is a fresh start. Give the answer in the form the question asks for.

1. A tennis tournament has 34 players from NSW, 20 from Victoria and 6 from Queensland. One player is picked at random. Find the probability that the player is from NSW.

Answer

2. Use the tournament in Question 1. Find the probability that the player is not from Victoria.

Answer

3. A raffle sells 5000 tickets and Ben buys 2. Find the probability that Ben wins. Write it as a decimal.

Answer

4. Write $\frac{7}{15}$ as a decimal, correct to 2 decimal places. Year 7

Answer

5. The spinner is spun once. Find the probability of not landing on yellow. Write it as a percentage.



Answer

6. A fair die shows a 6 five times in a row. Mai says a 6 is now more likely. Rani says a 6 is now less likely. Who is right? Explain.



Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. 3 PROBLEMS

1. 8K has 28 students: 12 catch the bus, 9 walk and the rest are driven to school. One student is picked at random. Find (a) P(driven) (b) P(does not walk).

Answer

2. The letters of TWEED HEADS are written on 10 cards. One card is picked without looking. Find (a) P(an E) (b) P(a vowel).

Answer

3. A charity raffle sells 2500 tickets at $2 each, with one prize. Lucy spends $20 on tickets. How many tickets does Lucy have? Find the probability that Lucy wins, as a percentage.

Answer

5 Observed probability

Probability • MA4-PRO-C-01

EASY •

LEARNING INTENTION

We are learning to find observed probabilities from experiments and data, and compare them with theoretical probabilities.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- ▸ *I can* find an observed probability: frequency ÷ total trials. ○○○
- ▸ *I can* tell theoretical and observed probability apart. ○○○
- ▸ *I can* run a simulation with a random number generator. ○○○

NOTES key ideas: students copy these into their booklet

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KEY TERMS

theoretical probability: from equally likely outcomes, e.g. P(6) = ⅙

observed probability (relative frequency): from the results of trials

frequency: how many times it happened

biased: unfair; some outcomes are more likely

simulation: copying an experiment, e.g. with random numbers

FORMULA

$$\text{observed P(E)} = \frac{\text{number of times E happened}}{\text{total number of trials}}$$

More trials: observed P is usually closer to theoretical P.

Calculator **RanInt#(1,6)**; spreadsheet **=RANDBETWEEN(1,6)**

EXAMPLE

Harper rolls a die 60 times; a 2 comes up 15 times.

observed $P(2) = \frac{15}{60} = \frac{1}{4}$

theoretical $P(2) = \frac{1}{6}$

$\frac{1}{4} > \frac{1}{6}$: more 2s than expected.

Biased? Roll many more times to check.

WE DO with your teacher: copy every step into the space

Example 1 Find each probability. Observed probability = the number of times it happened ÷ the total number of trials.

a Observed

A die is rolled 60 times and a 2 comes up 15 times. Find the observed probability of a 2.



b Compare

Find the theoretical probability of a 2 on a fair die. Is the observed probability in part a higher or lower?

c Spinner

Vinh spins the spinner 80 times: red 44, green 25, yellow 11. Find the observed probability of red.



d Percentage

Sefina tosses a coin 250 times: 135 heads and 115 tails. Find the observed probability of tails, as a percentage.



✎ YOU DO on your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Observed probability = $\frac{\text{number of times it happened}}{\text{total number of trials}}$. Theoretical probability comes from equally likely outcomes; observed probability comes from results.

ROUND 1 Questions 1–4: find each observed probability: the number of times it happened ÷ the number of trials. Write it in simplest form.

1 A coin is tossed 20 times and lands heads 12 times. Find the observed probability of heads.

Working _____

P = _____

2 A die is rolled 30 times and shows a 6 four times. Find the observed probability of a 6.

Working _____

P = _____

3 A basketballer takes 50 shots and scores 35. Find the observed probability of scoring.

Working _____

P = _____

4 A spinner is spun 25 times and lands on blue 10 times. Find the observed probability of blue.

Working _____

P = _____

ROUND 2 Questions 5–8: decide whether each probability is theoretical (T: worked out from equally likely outcomes) or observed (O: worked out from results). Write T or O.

5 $P(\text{heads}) = \frac{1}{2}$, because a coin has two equal sides. T or O?



T or O _____

6 Kingscliff won 14 of its 20 games, so P(win) = 0.7. T or O?

T or O _____

7 $P(\text{rolling a 3}) = \frac{1}{6}$ for a fair die. T or O?



T or O _____

8 120 of last year's 365 days were rainy, so P(rain) ≈ 0.33. T or O?

T or O _____

5 Observed probability

Probability • MA4-PRO-C-01

MEDIUM ••

YOU DO on your own: keep going, one round at a time. Show your working. **11 QUESTIONS**

REMEMBER The total number of trials goes on the bottom of the fraction. Add the frequencies to find the total. For '5 or 6', add their frequencies.

ROUND 3 Questions 9–11: Vinh spun this spinner 80 times. Use the results in the table to find each probability.



Colour	Red	Green	Yellow
Frequency	44	25	11

9 Find the observed probability of red.

Working _____

Answer _____

10 Find the theoretical probability of red. (Look at the spinner.)

Working _____

Answer _____

11 Find the observed probability of yellow.

Working _____

Answer _____

ROUND 4 Questions 12–14: the matches in each of 50 matchboxes were counted. Use the table.

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

12 How many matchboxes were tested? Check by adding the bottom row of the table.

Working _____

Answer _____

13 Find the observed probability that a box has exactly 51 matches.

Working _____

Answer _____

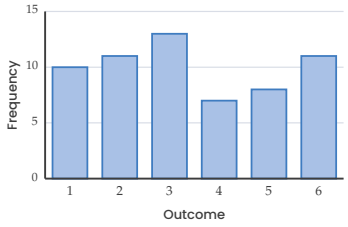
14 Find the observed probability that a box has fewer than 50 matches.

Working _____

Answer _____

ROUND 5 Questions 15–17: Declan rolled a die 60 times. Use the table and the graph.

Outcome	1	2	3	4	5	6
Frequency	10	11	13	7	8	11



15 Find the observed probability of a 3.

Working _____

Answer _____

16 Find the observed probability of a 5 or a 6.

Working _____

Answer _____

17 Find the theoretical probability of a 5 or a 6 on a fair die.

Working _____

Answer _____

ROUND 6 Questions 18–19: run your own experiment with a random number generator.

18 Use RanInt#(1,6) on your calculator (or =RANDBETWEEN(1,6) in a spreadsheet) to roll a die 30 times. Tally your results.

Outcome	1	2	3	4	5	6
Tally						
Frequency						

19 Find your observed probability of a 6. Compare it with the theoretical probability, $\frac{1}{6}$. Why might they be different?



Answer _____

5 Observed probability

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Use the tables. The total number of trials goes on the bottom of the fraction.

Matches in 50 boxes

Matches	48	49	50	51	52	53
Boxes	3	6	20	16	4	1

A die rolled 80 times

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

a Table

Use the matchbox table. Find the observed probability that a box has exactly 51 matches.

b Fewer than

Use the matchbox table. Find the observed probability that a box has fewer than 50 matches.

c Biased?

Use the table of 80 rolls. Do you think the die is biased (unfair)? Give a reason using the numbers.

d Simulate

Use RanInt#(1,6) on a calculator to roll a die 20 times. Record the results, then find your observed probability of an even number.



✎ YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. — 5 QUESTIONS

REMEMBER Observed probability = frequency ÷ total trials. More trials give a better estimate.

1 Kai tosses a coin 200 times and gets 92 heads. Find the observed probability of tails. Write it as a decimal.

Answer

2 Two dice were rolled 50 times and the sum of the two numbers was recorded each time. Find the observed probability of a sum of 8, as a decimal.

Sum	2	3	4	5	6	7	8	9	10	11	12
Frequency	0	2	4	6	5	5	9	6	8	3	2

Answer

3 Use the table in Question 2. Find the observed probability of a sum of 5 or 6.

Answer

4 Use the table in Question 2. Which sum came up most often? Find its observed probability.

Answer

5 Write $\frac{13}{50}$ as a percentage. Year 7

Answer

✎ WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. — 3 PROBLEMS

1 Harper rolled a die 60 times and 2 came up 15 times. Harper says the die is biased. Is there enough evidence? What could Harper do to check?

Answer

2 The canteen's last 200 lunch orders were 70 wraps, 50 sushi and 80 burgers. Find the observed probability that the next order is a wrap. Why is this only an estimate?

Answer

3 To estimate the chance that a family of 4 children are all girls, Jo tosses 4 coins together (heads = girl), 50 times. All 4 coins showed heads 4 times. Find the observed probability. How could Jo improve the estimate?



Answer

6 Expected frequency

Probability • MA4-PRO-C-01 EASY •

LEARNING INTENTION

We are learning to calculate the expected frequency of an event, and compare it with the observed frequency.

SUCCESS CRITERIA rate yourself at the end: ○○○ not yet · getting there · got it

- *I can* find expected frequency = probability × number of trials. ○○○
- *I can* compare observed and expected frequencies. ○○○
- *I can* use an observed probability to make a prediction. ○○○

NOTES key ideas: students copy these into their booklet TEACHER COPY

KEY TERMS

expected frequency: how many times we expect an event in a number of trials

observed frequency: how many times it actually happened

fair: every outcome equally likely; **biased:** not fair

Observed is rarely exactly expected. A big gap over many trials suggests bias.

FORMULA

expected frequency = probability × number of trials

From data, use the observed P: 15 twos in 60 rolls, so $\frac{1}{4} \times 500 = 125$.

Not always whole: $\frac{1}{6} \times 80 \approx 13.3$

EXAMPLE

Declan rolls a die 60 times. Expected 5s or 6s:

$P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$

expected $= \frac{1}{3} \times 60 = 20$

observed $= 8 + 11 = 19$: close to 20, so the die seems fair.

WE DO with your teacher: copy every step into the space

Example 1 Find each expected frequency: the probability × the number of trials.

a Coin

A coin is tossed 50 times. Find the expected number of heads.



b Die

A die is rolled 72 times. Find the expected number of 4s.



c Spinner

Vinh spins the spinner 80 times. Find the expected number of reds. He actually got 44 reds: compare.



d Rain

The probability of rain on an April day is 30%. April has 30 days. Find the expected number of rainy days.



YOU DO on your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER Expected frequency = probability × number of trials. For example, 6s in 60 rolls: $\frac{1}{6} \times 60 = 10$.

ROUND 1 Questions 1–4: find each expected frequency. Write the probability, multiply by the number of trials, then write the expected number.

1 A coin is tossed 100 times. How many heads would you expect?



Probability _____

× number of trials _____

Expected _____

2 A die is rolled 60 times. How many 6s would you expect?



Probability _____

× number of trials _____

Expected _____

3 This spinner is spun 40 times. How many blues would you expect?



Probability _____

× number of trials _____

Expected _____

4 A die is rolled 30 times. How many even numbers would you expect?



Probability _____

× number of trials _____

Expected _____

ROUND 2 Questions 5–8: Vinh's spinner (half red, a quarter green, a quarter yellow) was spun 80 times: red 44, green 25, yellow 11.

5 How many reds would you expect in 80 spins?

Working _____

Answer _____

6 How many yellows would you expect in 80 spins?

Working _____

Answer _____

7 Red actually came up 44 times. Is that more or less than expected? By how many?

Working _____

Answer _____

8 Yellow actually came up 11 times. Is that more or less than expected? By how many?

Working _____

Answer _____

6 Expected frequency

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MEDIUM ●●

✎ YOU DO — on your own: keep going, one round at a time. Show your working. — 11 QUESTIONS

REMEMBER Expected frequency = probability × number of trials. From data, use the observed probability. The answer might not be a whole number.

ROUND 3 Questions 9–11: a fair die is rolled 72 times. Find each expected frequency: probability × 72.

9 How many 4s would you expect?

Working _____

Answer _____

10 How many numbers greater than 2 would you expect?

Working _____

Answer _____

11 How many even numbers would you expect?

Working _____

Answer _____

ROUND 4 Questions 12–14: use the observed probability (from what happened before) to predict what will happen.

12 A basketballer scores 7 of every 10 shots. How many goals would you expect from 50 shots?

Working _____

Answer _____

13 A surfer catches a wave on 3 of every 4 attempts. How many waves would you expect from 60 attempts?

Working _____

Answer _____

14 In a sample, 20 of 50 matchboxes held exactly 50 matches. How many of 250 boxes would you expect to hold exactly 50?

Working _____

Answer _____

ROUND 5 Questions 15–17: Kingscliff weather records show that the probability of a summer day above 30 °C is 0.2.

15 How many days above 30 °C would you expect in a 90-day summer?

Working _____

Answer _____

16 How many days above 30 °C would you expect in two weeks (14 days)?

Working _____

Answer _____

17 Give a reason why the real number of hot days might be different.

Working _____

Answer _____

ROUND 6 Questions 18–19: do a coin experiment, then compare what you observed with what you expected.

18 Toss a coin 50 times (or use RanInt#(1,2), 1 = heads). Tally the results.

Outcome	Tally	Frequency
Heads		
Tails		
Total		50

19 How many heads would you expect in 50 tosses? Compare this with the number of heads you actually got.

Answer _____

6 Expected frequency

Probability • MA4-PRO-C-01

HARD •••

WE DO with your teacher: copy every step into the space

Example 2 Expected = probability × trials. If the outcomes are not equally likely, use the observed probability from the data.

a From data

Harper's die gave 15 twos in 60 rolls. Use this observed probability to find how many 2s to expect in 500 rolls.



b Matchboxes

In a sample, 20 of 50 matchboxes held exactly 50 matches. How many of 1000 boxes would you expect to hold exactly 50?

c Not whole

A die is rolled 80 times. Find the expected number of 6s (to 1 decimal place). A 6 actually came up 22 times. What does this suggest?

d Lollies

For a bag of lollies, P(red) = 0.5, P(green) = 0.2 and P(purple) = 0.3. How many of each colour would you expect in a bag of 100 lollies?

YOU DO on your own: work in order. A tag shows an earlier lesson or year. Show your working. **5 QUESTIONS**

REMEMBER Expected = probability × trials. If observed is close to expected, the die or spinner seems fair; if it is far off over many trials, it may be biased.

1 A die was rolled 80 times (see the table). Find the expected number of 6s, to 1 decimal place. Compare it with the number of 6s that actually came up.

Outcome	1	2	3	4	5	6
Frequency	11	13	9	13	12	22

Answer

2 Use Question 1. Do you think the die is biased? Explain.

Answer

3 A fair die is rolled 100 times. Find the expected number of 3s, to 1 decimal place.

Answer

4 Find 35% of 240. Year 7

Answer

5 In a sample, 41 of 50 matchboxes held at least 50 matches. How many of 1000 boxes would you expect to hold at least 50?

Answer

WORDED PROBLEMS draw a diagram or a table if it helps, then solve. Show your working. **3 PROBLEMS**

1 Declan rolled a die 60 times: 1 (10), 2 (11), 3 (13), 4 (7), 5 (8), 6 (11). Find the expected number of 5s or 6s and the observed number of 5s or 6s. Is the die fair? Explain.

Answer

2 A spinner has 5 equal sectors: 2 red, 2 blue, 1 green. Mel says she will get exactly 40 reds in 100 spins. Explain why she might not.

Answer

3 8 students each toss a coin 100 times. How many heads would you expect altogether? The class got 389 heads. Is that close? Explain.

Answer

CHAPTER SUMMARY

the key facts from every lesson, and how confident you are with each

HOW CONFIDENT AM I?

○○○ not yet · getting there · got it

1	Chance and the sample space	○○○	2	Probability of an event	○○○
3	Complementary events	○○○	4	Probability problems	○○○
5	Observed probability	○○○	6	Expected frequency	○○○

Key facts

Sample space: every possible outcome, each written once, in braces { } (L1)

$\mathbf{P(E)} = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$ when outcomes are equally likely (L2)

'or' for different groups: add the counts, e.g. P(red or blue) (L2)

Checklist: total → favourable → 'not'? → form asked → check (L4)

More trials: the observed probability usually gets closer to the theoretical one (L5)

Fair or biased? Close to expected: probably fair. Far off over many trials: probably biased (L6)

Probability scale: 0 impossible · $\frac{1}{2}$ even chance · 1 certain (L1)

Converting: fraction → decimal: top ÷ bottom; decimal → percentage: × 100% (L2)

Complement: P(not E) = 1 − P(E); all probabilities add to 1 (L3)

Observed P(E) $= \frac{\text{number of times E happened}}{\text{total number of trials}}$ (L5)

Expected frequency = probability × number of trials (L6)

IF YOU WERE AWAY

two examples from each lesson, fully worked: read them, then try the lesson's first round

Lesson 1 • Chance and the sample space

the lesson is on p. 2

Example 1: a sample space

List the sample space for a spinner with sectors red, blue, red, green.

Step 1 Write each colour once: {red, blue, green}.

Step 2 Count: 3 outcomes.

Step 3 Red has 2 of the 4 sectors, so the outcomes are **not equally likely**.

Example 2: chance words

The forecast says a 30% chance of rain. Describe the chance.

Step 1 Compare with 50% (even chance): 30% is less.

Step 2 So rain is **unlikely**.

Step 3 It is not 0%, so it is still possible.

Lesson 2 • Probability of an event

the lesson is on p. 5

Example 1: P(E) from counters

A bag has 5 red, 3 blue and 4 green counters. Find P(red).

Step 1 Total = 5 + 3 + 4 = 12.

Step 2 Favourable: 5 red.

Step 3 P(red) $= \frac{5}{12}$

Example 2: a percentage

A die is rolled. Find P(a number greater than 4), as a percentage.

Step 1 Favourable: {5, 6}, 2 of 6.

Step 2 $\frac{2}{6} = \frac{1}{3}$

Step 3 $\frac{1}{3} \times 100\% = 33\frac{1}{3}\% \approx 33.3\%$

Lesson 3 • Complementary events

the lesson is on p. 8

Example 1: P(not E) with decimals

$P(\text{rain}) = 0.15$. Find $P(\text{no rain})$.

Step 1 'No rain' is the complement of 'rain'.

Step 2 $\text{P}(\text{no rain}) = 1 - 0.15 = \mathbf{0.85}$

Example 2: P(not E) with fractions

A pack has 10 red, 6 yellow and 4 green cards. Find $P(\text{not yellow})$.

Step 1 $\text{P}(\text{yellow}) = \frac{6}{20} = \frac{3}{10}$

Step 2 $\text{P}(\text{not yellow}) = 1 - \frac{3}{10} = \mathbf{\frac{7}{10}}$

Lesson 4 • Probability problems

the lesson is on p. 11

Example 1: an 'or' problem

A number from 1 to 15 is chosen. Find $P(\text{a multiple of 3 or 5})$.

Step 1 List: $\{3, 5, 6, 9, 10, 12, 15\}$ (15 once).

Step 2 7 of 15: $\text{P} = \mathbf{\frac{7}{15}}$

Example 2: no memory

A coin has shown tails 7 times in a row. Find $P(\text{tails next})$.

Each toss is a fresh start: the coin does not remember.

$\text{P}(\text{tails}) = \mathbf{\frac{1}{2}}$

Lesson 5 • Observed probability

the lesson is on p. 14

Example 1: observed probability

A die is rolled 60 times; 2 comes up 15 times. Find the observed $P(2)$.

Step 1 Observed P = times ÷ trials $= \frac{15}{60} = \mathbf{\frac{1}{4}}$

Step 2 Theoretical $\text{P}(2) = \frac{1}{6}$, so 2 came up more than expected.

Example 2: from a table

Matches in 50 boxes: 48 (3), 49 (6), 50 (20), 51 (16), 52 (4), 53 (1). Find the observed $P(\text{fewer than } 50)$.

Step 1 Fewer than 50: 48 or 49, $3 + 6 = 9$ boxes.

Step 2 $\text{P} = \mathbf{\frac{9}{50}}$

Lesson 6 • Expected frequency

the lesson is on p. 17

Example 1: expected frequency

A die is rolled 72 times. Find the expected number of 4s.

Expected = P × trials $= \frac{1}{6} \times 72 = \mathbf{12}$

Example 2: predicting from data

Harper's die gave 15 twos in 60 rolls. How many 2s in 500 rolls?

Step 1 Observed $\text{P}(2) = \frac{15}{60} = \frac{1}{4}$

Step 2 $\frac{1}{4} \times 500 = \mathbf{125}$

R1 [L1] List the sample space for a spinner with sectors 1, 2, 2 and 3. How many outcomes are there?

R2 [L1] A drawing pin is dropped: it lands point up or point down. Are the outcomes equally likely? Explain.

R3 [L2] A bag has 6 red, 4 blue and 2 white marbles. One is taken without looking. Find the probability of blue.

R4 [L2] A die is rolled. Find the probability of a number less than 5, as a percentage.

R5 [L2] One card is taken from a standard deck. Find the probability of a black king.

R6 [L2] Write $\frac{7}{20}$ as a decimal and as a percentage.

R7 [L3] Write the complement of ‘Kingscliff wins the final’.

R8 [L3] The probability that the bus is late is 0.12. Find the probability that it is not late.

R9 [L3] A spinner is red, blue or green. $P(\text{red}) = 0.45$ and $P(\text{blue}) = 0.3$. Find $P(\text{green})$.

R10 [L3] A family of 5 (2 parents, 3 children) take turns to wash up. Find $P(\text{the washer-up is not a parent})$.

R11 [L4] One tile is picked from T, W, E, E, D. Find the probability of an E, as a decimal.

R12 [L4] The digits 1, 3, 5 and 8 make a 4-digit number at random, each used once. Find P(the number is odd).

R13 [L5] A coin is tossed 80 times and lands heads 36 times. Find the observed probability of heads, as a decimal.

R14 [L5] Theoretical or observed? 'P(a Year 8 student surfs) = 0.3, from a survey.' Explain.

R15 [L6] A die is rolled 90 times. How many 5s would you expect?

R16 [L6] A basketballer scored 30 of 50 shots. Predict the number scored in 120 shots.

EXIT TICKETS cut along the dashed lines • write your summary on the back • hand it to your teacher

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EXIT TICKET Lesson 1: Chance and the sample space Name _____ Class _____

Easy • A spinner has 3 equal sectors: red, blue, red. List the sample space.

Medium •• On that spinner, are red and blue equally likely? Explain.

Hard ••• A die is rolled. Write the chance word for rolling a number less than 3.

EXIT TICKET Lesson 2: Probability of an event Name _____ Class _____

Easy • A die is rolled. Find the probability of a 4.

Medium •• A bag has 3 red and 5 blue counters. Find the probability of blue, as a percentage.

Hard ••• One card is taken from a deck. Find the probability of a red ace.

EXIT TICKET Lesson 3: Complementary events Name _____ Class _____

Easy • Write the complement of 'rolling an even number'.

Medium •• The probability of winning is 0.35. Find the probability of not winning.

Hard ••• A jar has 7 red, 5 green and 8 yellow lollies. One is taken. Find the probability it is not yellow.

EXIT TICKET Lesson 4: Probability problems Name _____ Class _____

Easy • A spinner has 5 equal sectors numbered 1 to 5. Find the probability of an odd number, as a percentage.

Medium •• Of 30 students, 12 catch the bus, 10 walk and the rest are driven. Find P(driven).

Hard ••• A coin lands heads 5 times in a row. Find the probability of heads on the next toss. Explain.

✂

SUMMARISE LESSON 1 IN YOUR OWN WORDS

Chance and the sample space • the back of your exit ticket

✂

SUMMARISE LESSON 2 IN YOUR OWN WORDS

Probability of an event • the back of your exit ticket

✂

SUMMARISE LESSON 3 IN YOUR OWN WORDS

Complementary events • the back of your exit ticket

✂

SUMMARISE LESSON 4 IN YOUR OWN WORDS

Probability problems • the back of your exit ticket

✂

✂ - - -

✂

SUMMARISE LESSON 5 IN YOUR OWN WORDS

Observed probability • the back of your exit ticket

✂

SUMMARISE LESSON 6 IN YOUR OWN WORDS

Expected frequency • the back of your exit ticket

✂

HOMEWORK STRIPS after each lesson, cut out its strip (start at the top right), do it at home, and hand it in next lesson

HW 7 Before the test 1 ✂

1 **L1** How many outcomes are in the sample space for the day of the week?

2 **L2** A 10-sided die (1 to 10) is rolled. Find the probability of an even number.

3 **L3** A die is rolled. Find the probability of not rolling a 3.

HW 4 L4 Problems ✂

1 A whole number from 1 to 10 is chosen at random. Find the probability of a prime number.

2 One tile is picked from T, W, E, E, D. Find the probability that it is not an E.

3 A bag of 20 lollies has P(red) = 0.5. How many lollies are red?

HW 1 L1 Chance ✂

1 A coin is tossed. List the sample space.

2 An event has a probability of 0.8. Write its chance word.

3 A fair die is rolled. Are the outcomes 1 to 6 equally likely?

HW 8 Before the test 2 ✂

1 **L4** A drawer has 3 black socks and 1 white sock. One sock is taken. Find P(white).

2 **L5** A spinner landed on blue in 12 of 40 spins. Find the observed probability of blue.

3 **L6** P(E) = 0.3. How many times would you expect E in 50 trials?

HW 5 L5 Observed ✂

1 A coin is tossed 50 times and lands heads 27 times. Find the observed probability of heads.

2 Is 'P(heads) = ½' a theoretical or an observed probability?

3 Why is it better to do many trials?

HW 2 L2 P(E) ✂

1 A die is rolled. Find the probability of a number greater than 2.

2 A bag has 2 red and 6 blue counters. Find the probability of red, as a decimal.

3 One card is taken from a deck. Find the probability of a spade.

HW 9 Before the test 3 ✂

1 **L2** One card is taken from a deck. Find the probability of an ace.

2 **L3** The probability of winning is 65%. Find the probability of not winning.

3 **L6** A die rolled 60 times gave 25 sixes. Do you think it is biased? Explain.

HW 6 L6 Expected ✂

1 A die is rolled 300 times. How many 6s would you expect?

2 A coin is tossed 70 times. How many heads would you expect?

3 A spinner is half red. It is spun 40 times. How many reds would you expect?

HW 3 L3 Complements ✂

1 The probability of rain is 0.4. Find the probability of no rain.

2 Write the complement of 'winning a raffle'.

3 One tile is picked from N, I, P, P, E, R, S. Find the probability that it is not a vowel.

HW 1 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 2 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 3 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 4 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 5 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 6 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 7 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 8 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

HW 9 Name and working

Name

Class Due

How did it go? ○○○ easy · okay · stuck

More working:

Lesson 1 · Chance and the sample space

PAGES 1–2 · EASY AND MEDIUM ROUNDS

1 0 of 3 sectors: impossible
2 1 of 5 sectors: unlikely
3 2 of 4 sectors: even chance
4 4 of 6 sectors: likely
5 {heads, tails}: 2 outcomes
6 {1, 2, 3, 4, 5, 6}: 6 outcomes
7 {2, 4, 6, 8}: 4 outcomes
8 {T, W, E, D}: 4 outcomes (E is written once)
9 {red, blue, green}: 3 outcomes
10 {J, F, M, A, S, O, N, D}: 8 outcomes
11 {win, lose, draw}: 3 outcomes
12 {1, 3, 5}
13 {U, E}
14 {5, 6, 7, 8, 9}
15 No. Red covers half the spinner; blue and green cover a quarter each.
16 Yes. Each face of a fair die has the same chance.
17 No. It depends on how well the person drives.
18 likely
19 even chance
20 unlikely (almost impossible)

PAGE 3 · HARD

1 A between 0 and ½ (near 0), B at 1, C at ½.
2 0.75 and 75%
3 {K, I, N, G, S, C, L, F}: 8 outcomes
4 Red: it has 3 sectors and blue has only 2.
5 No. On a fair die every number is equally likely.
6 No. It is still an even chance: the coin does not remember earlier tosses.

WORDED PROBLEMS

W1 Likely. It is not certain, because 70% is less than 100%.
W2 3 blue sectors and 1 other colour: 3 of 4 is more than half but less than all.
W3 {Amy, Ben, Cara, Dev}. No: 2 are from each class, so it is an even chance.

Lesson 2 · Probability of an event

PAGES 1–2 · EASY AND MEDIUM ROUNDS

1 $\frac{1}{2}$
2 $\frac{1}{3}$
3 $\frac{2}{5}$
4 $\frac{3}{8}$
5 {5}: $\frac{1}{6}$
6 {2, 4, 6}: $\frac{3}{6} = \frac{1}{2}$
7 {1, 2}: $\frac{2}{6} = \frac{1}{3}$
8 {1, 2, 3, 6}: $\frac{4}{6} = \frac{2}{3}$
9 $\frac{5}{12}$
10 $\frac{3}{12} = \frac{1}{4}$
11 $\frac{9}{12} = \frac{3}{4}$
12 $\frac{1}{2} = 0.5 = 50\%$
13 $\frac{1}{4} = 0.25 = 25\%$
14 $\frac{2}{10} = \frac{1}{5} = 0.2 = 20\%$
15 $\frac{4}{52} = \frac{1}{13}$
16 $\frac{13}{52} = \frac{1}{4}$
17 $\frac{26}{52} = \frac{1}{2}$
18 $\frac{3}{20} = 0.15$
19 $\frac{9}{20} = 45\%$
20 $\frac{8}{20} = \frac{2}{5}$

PAGE 3 · HARD

1 $\frac{9}{16}$
2 0.375 and 37.5%
3 {2, 3, 5, 7, 11, 13}: $\frac{6}{15} = \frac{2}{5}$
4 $\frac{3}{7}$
5 {3, 4, 5, 6}: $\frac{4}{6} = \frac{2}{3}$
6 $\frac{8}{500} = \frac{2}{125} = 1.6\%$

WORDED PROBLEMS

W1 $\frac{8}{15}$. Add 1 cola: then 8 of the 16 cans are lemonade.
W2 $P(7) = \frac{8}{20} = \frac{2}{5} = 0.4 = 40\%$. $P(\text{even}) = \frac{7}{20} = 0.35 = 35\%$.
W3 $30\% \times 30 = 9$ days

Lesson 3 · Complementary events

PAGES 1–2 · EASY AND MEDIUM ROUNDS

1 Tossing heads
2 Not rolling a 6 (rolling 1, 2, 3, 4 or 5)
3 Not a cloudy day tomorrow
4 Choosing a dark chocolate
5 $P(\text{blue}) = \frac{1}{4}$, $P(\text{not blue}) = \frac{3}{4}$; $\frac{1}{4} + \frac{3}{4} = 1$
6 $P(\text{blue}) = \frac{2}{3}$, $P(\text{not blue}) = \frac{1}{3}$; $\frac{2}{3} + \frac{1}{3} = 1$
7 $P(\text{blue}) = \frac{1}{5}$, $P(\text{not blue}) = \frac{4}{5}$; $\frac{1}{5} + \frac{4}{5} = 1$
8 $P(\text{blue}) = \frac{1}{2}$, $P(\text{not blue}) = \frac{1}{2}$; $\frac{1}{2} + \frac{1}{2} = 1$
9 {1, 2, 4, 5, 6}: $\frac{5}{6}$
10 {1, 3, 5}: $\frac{3}{6} = \frac{1}{2}$
11 {1, 4, 6} (1 is not prime): $\frac{3}{6} = \frac{1}{2}$
12 $100\% - 85\% = 15\%$
13 $1 - 0.1 = 0.9$
14 $1 - \frac{1}{13} = \frac{12}{13}$
15 $\frac{2}{6} = \frac{1}{3}$
16 $1 - \frac{1}{3} = \frac{2}{3}$
17 1 (there are no science books, so it is certain)
18 E: {3, 6, 9}; not E: {1, 2, 4, 5, 7, 8, 10}
19 $P(\text{E}) = \frac{3}{10}$, $P(\text{not E}) = \frac{7}{10}$
20 $1 - 0.6 - 0.1 = 0.3$

PAGE 3 · HARD

1 $1 - \frac{450}{560} = \frac{110}{560} = \frac{11}{56}$

2 $\frac{5}{8}$, 0.63, 36%

3 $1 - \frac{15}{130} = \frac{115}{130} = \frac{23}{26}$

4 blue or black = 75, so not = $\frac{55}{130} = \frac{11}{26}$

5 $\frac{8}{10} = \frac{4}{5} = 0.8$

6 7 consonants: $\frac{7}{11}$

WORDED PROBLEMS

W1 P(blue) = 0.35. Red 16, white 10, blue 14.

W2 P(win) = $\frac{7}{25}$, P(not win) = $\frac{18}{25}$

W3 $\frac{3}{4}$, $\frac{1}{2}$, 0

Lesson 4 · Probability problems

PAGES 1–2 · EASY AND MEDIUM ROUNDS

1 $\frac{1}{100}$

2 $\frac{60}{100} = \frac{3}{5}$

3 $\frac{50}{100} = \frac{1}{2}$

4 $\frac{25}{100} = \frac{1}{4}$

5 $\frac{1}{5} = 20\%$

6 $\frac{2}{5} = 40\%$

7 $\frac{4}{5} = 80\%$

8 $\frac{5}{5} = 1 = 100\%$ (certain)

9 $\frac{17}{28}$

10 $\frac{24}{28} = \frac{6}{7}$

11 $\frac{6}{28} = \frac{3}{14}$

12 {black, blue, red}

13 $\frac{2}{6} = \frac{1}{3}$

14 0 (impossible)

15 {1, 3, 5, 7, 9, 11, 13, 15}: $\frac{8}{15}$

16 {1, 4, 6, 8, 9, 10, 12, 14, 15}: $\frac{9}{15} = \frac{3}{5}$

17 {1, 2, 3, 4, 6, 12}: $\frac{6}{15} = \frac{2}{5}$

18 $\frac{2}{7}$

19 $\frac{5}{7}$

20 0: no letter of SUCCESS is in FAIL

PAGE 3 · HARD

1 $\frac{34}{60} = \frac{17}{30}$

2 $\frac{40}{60} = \frac{2}{3}$

3 $\frac{2}{5000} = \frac{1}{2500} = 0.0004$

4 0.47

5 $\frac{6}{8} = \frac{3}{4} = 75\%$

6 Neither: P(6) = $\frac{1}{6}$ on every roll of a fair die.

WORDED PROBLEMS

W1 Driven 7: $\frac{7}{28} = \frac{1}{4}$. Not walk: $\frac{19}{28}$.

W2 P(E) = $\frac{3}{10}$; P(vowel) = $\frac{4}{10} = \frac{2}{5}$

W3 10 tickets: $\frac{10}{2500} = \frac{1}{250} = 0.4\%$

Lesson 5 · Observed probability

PAGES 1–2 · EASY AND MEDIUM ROUNDS

1 $\frac{12}{20} = \frac{3}{5}$

2 $\frac{4}{30} = \frac{2}{15}$

3 $\frac{35}{50} = \frac{7}{10}$

4 $\frac{10}{25} = \frac{2}{5}$

5 T

6 O

7 T

8 O

9 $\frac{44}{80} = \frac{11}{20}$

10 $\frac{1}{2}$ (red is half the spinner)

11 $\frac{11}{80}$

12 3 + 6 + 20 + 16 + 4 + 1 = 50

13 $\frac{16}{50} = \frac{8}{25}$

14 $\frac{9}{50}$

15 $\frac{13}{60}$

16 $\frac{19}{60}$

17 $\frac{2}{6} = \frac{1}{3}$

18 Answers vary (frequencies should add to 30).

19 Answers vary. With only 30 trials the observed probability can be quite far from ⅙.

PAGE 3 · HARD

1 $\frac{108}{200} = \frac{27}{50} = 0.54$

2 $\frac{9}{50} = 0.18$

3 $\frac{11}{50}$

4 8: $\frac{9}{50}$

5 26%

WORDED PROBLEMS

W1 Observed P(2) = ¼, but a fair die gives about 10 twos in 60 rolls. 15 is suspicious but 60 rolls is not many: roll it hundreds more times.

W2 $\frac{70}{200} = \frac{7}{20} = 0.35$. It is based on past orders; tomorrow's orders may be different.

W3 $\frac{4}{50} = \frac{2}{25} = 0.08$. Run many more trials.

Lesson 6 · Expected frequency

PAGES 1–2 · EASY AND MEDIUM ROUNDS

1 $\frac{1}{2} \times 100 = 50$

2 $\frac{1}{6} \times 60 = 10$

3 $\frac{1}{4} \times 40 = 10$

4 $\frac{1}{2} \times 30 = 15$

5 $\frac{1}{2} \times 80 = 40$

6 $\frac{1}{4} \times 80 = 20$

7 More, by 4
8 9 fewer than expected
9 $\frac{1}{6} \times 72 = 12$
10 $\frac{2}{3} \times 72 = 48$
11 $\frac{1}{2} \times 72 = 36$
12 $\frac{7}{10} \times 50 = 35$
13 $\frac{3}{4} \times 60 = 45$
14 $\frac{2}{5} \times 250 = 100$
15 $0.2 \times 90 = 18$
16 $0.2 \times 14 = 2.8$
17 Chance: the weather varies from year to year; 0.2 is only an estimate.
18 Answers vary.
19 $\frac{1}{2} \times 50 = 25$; comparison varies.

PAGE 3 · HARD

1 $\frac{1}{6} \times 80 \approx 13.3$; 22 is much higher.
2 Probably: 6 came up far more often than expected (22 compared with 13.3).
3 $\frac{1}{6} \times 100 \approx 16.7$
4 84
5 $\frac{41}{50} \times 1000 = 820$

WORDED PROBLEMS

W1 Expected $\frac{1}{3} \times 60 = 20$; observed $8 + 11 = 19$. Close, so the die seems fair.
W2 40 is the expected number ($\frac{2}{5} \times 100 = 40$), but results vary by chance.
W3 $\frac{1}{2} \times 800 = 400$. 389 is close (observed P = 0.486…).

Chapter review

R1 {1, 2, 3}: 3 outcomes
R2 No: the pin is not symmetrical, so one way is more likely.
R3 $\frac{4}{12} = \frac{1}{3}$
R4 $\frac{4}{6} = \frac{2}{3} = 66\frac{2}{3}\% \approx 66.7\%$
R5 $\frac{2}{52} = \frac{1}{26}$
R6 0.35 and 35%
R7 Kingscliff does not win (it loses or draws).
R8 0.88
R9 0.25
R10 $\frac{3}{5}$
R11 $\frac{2}{5} = 0.4$
R12 $\frac{3}{4}$ (it ends in 1, 3 or 5)
R13 $\frac{36}{80} = \frac{9}{20} = 0.45$
R14 Observed: it comes from survey results.
R15 15
R16 $\frac{3}{5} \times 120 = 72$

Exit tickets

L1 E {red, blue} · **M** No: red has 2 of the 3 sectors. · **H** Unlikely (2 of 6 outcomes).
L2 E $\frac{1}{6}$ · **M** $\frac{5}{8} = 62.5\%$ · **H** $\frac{2}{52} = \frac{1}{26}$
L3 E rolling an odd number · **M** 0.65 · **H** $\frac{12}{20} = \frac{3}{5}$
L4 E 60% · **M** $\frac{8}{30} = \frac{4}{15}$ · **H** $\frac{1}{2}$: the coin has no memory.
L5 E $\frac{18}{40} = \frac{9}{20}$ · **M** Theoretical · **H** Too few trials: chance has a big effect.
L6 E 30 · **M** 20 · **H** Probably: 34 is far more than the 20 expected.

Homework strips

HW 1 1 {heads, tails} · **2** likely · **3** Yes
HW 2 1 $\frac{4}{6} = \frac{2}{3}$ · **2** $\frac{2}{8} = \frac{1}{4} = 0.25$ · **3** $\frac{13}{52} = \frac{1}{4}$
HW 3 1 0.6 · **2** not winning the raffle · **3** $\frac{5}{7}$
HW 4 1 $\frac{4}{10} = \frac{2}{5}$ · **2** $\frac{3}{5}$ · **3** 10
HW 5 1 $\frac{27}{50}$ · **2** Theoretical · **3** The observed probability gets closer to the true one.
HW 6 1 50 · **2** 35 · **3** 20
HW 7 1 7 · **2** $\frac{5}{10} = \frac{1}{2}$ · **3** $\frac{5}{6}$
HW 8 1 $\frac{1}{4}$ · **2** $\frac{12}{40} = \frac{3}{10}$ · **3** 15
HW 9 1 $\frac{4}{52} = \frac{1}{13}$ · **2** 35% · **3** Probably: 10 sixes were expected.

TEXTBOOK COVERAGE New Century Maths 8, Chapter 8: where each exercise's skills appear (questions are rewritten, not copied)

Section	Textbook questions	Where in this booklet
8.01 Probability	Q1 spinners	L1 Ex 1b, Round 1; L2 Round 1
	Q2 count outcomes, equally likely	L1 Rounds 2, 5 (std); L1 Rounds 2, 4 (ext)
	Q3 list outcomes in events	L1 Round 4 (std), Round 3 (ext)
	Q4 money box	L2 Ex 2a; L2 Round 4 (ext)
	Q5 jar of lollies	L2 Ex 1c, Round 6 (std)
	Q6, Q8, Q10 multiple choice	L2 Round 6 Q20 (ext); L2 page 3 Q1; L2 Ex 2d
	Q7 match sample space	L1 Ex 2b
	Q9 Malik's cards	L2 worded W2 (std)
	Q11 values ↔ chance words	L1 Round 6 (std), Round 6 (ext)
	Q12 deck of cards	L2 Ex 2b, Round 5 (std), Round 3 (ext)
	Q13 rain in April	L2 W3 (std); L6 Ex 1d
	Q14 name from the roll	L1 W3 (std); L3 W3 (std)
	Q15 Esky	L2 W1 (std); L2 page 3 Q1–2 (ext)
	Investigation: complementary events	L3 notes and Ex 1
8.02 Complementary events	Q1 write complements	L3 Round 1 (std and ext)
	Q2–Q4 not red, die, books	L3 Ex 1b, Rounds 3 and 5 (std), Round 2 (ext)
	Q5 car park	L3 page 3 Q1 (std and ext)
	Q6 teacher's birth month	L1 Round 3 Q10 (months); L2 Round 6 (ext)
	Q7 phone number; Q8 raffle	L3 page 3 Q5 (std); L3 Ex 2d
	Q9 jelly beans; Q10 complement values	L3 page 3 Q3–4 (std), Rounds 3–4 (ext); L3 Round 4 (std)
	Q11 toy cars	L3 Ex 2b, W1 (std and ext)
	Q12 captain; Q13 rain 85%; Q14 complement of winning	L3 W3 (std); L3 Round 4; L3 Ex 2c
	Q15 coin grid game	L3 W2 (std), W3 (ext)
	Q16 PROBABILITY; Q17 washing up; Q18 4-digit numbers	L3 page 3 Q6 (std), Round 6 (ext); Review R10 (std); L2 W1 (ext), Review R12 (std)
8.03 Venn diagrams*	Q1–Q8	E1 (extension only): Ex 1, Rounds 1–6, page 3 Q1
8.04 Two-way tables*	Q1–Q7	E2 (extension only): Ex 1–2, Rounds 1–6, page 3 Q1–2
8.05 Probability problems	Q1–Q9, Q13	L4 Ex 1–2, Rounds 1–6 (std and ext); L3 Round 4 (eggs)
	Q10* tennis	L4 page 3 Q1–2 (std); L4 Round 3 (ext)
	Q11* camp Venn; Q12* hockey/judo	E1 page 3 Q2; E2 Round 6 (extension only)
8.06 Observed and expected	Q1 Harper's die	L5 Ex 1a–b; L6 Ex 2a
	Q2, Q5 coin and die experiments	L5 Round 6; L6 Ex 1b, Rounds 3 and 6
	Q3 Vinh's spinner	L5 Ex 1c, Round 3; L6 Ex 1c, Round 2
	Q4 matchboxes	L5 Ex 2a–b, Round 4; L6 Ex 2b, Round 4
	Q6 Sefina; Q7 die 80 rolls	L5 Ex 1d; L5 Ex 2c, L6 page 3 Q1–2
	Q8 dice sums	L5 page 3 Q2–4 (std), Round 5 (ext); L6 Round 6 (ext)
Technology and investigations	rolling a die, tossing a coin, family of 4	L5 Round 6, W3 (std), W1 (ext); L6 Round 6
Test yourself 8	Q1–Q6, Q10–Q11; Q7–Q9*	L1–L6 and review; E1 page 3 Q3, E2 Rounds 4–5

Not included Two-up (Did you know?) and lotto contexts (left out by choice), Power plus (not selected), Mental skills 8 (whole-number skills, not probability). * marks Year 9 path (Stage 5 Path MA5-PRO-P-01) content, which appears only in the extension booklet. The 2022 syllabus places 'at least / at most' wording in Stage 5 Path; the textbook uses it in 8.01–8.02, so the standard booklet tags those questions **Extension**.